

# STATISTICS

## Hypothesis Testing PART 2

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Rev: 03.12.2024

# Hypothesis Testing

## Statistics

There are four possible situations in the process of hypothesis testing.

| Decision     | Exact Situation                      |                                     |
|--------------|--------------------------------------|-------------------------------------|
|              | $H_0$ is True                        | $H_0$ is False                      |
| Accept $H_0$ | Correct Decision<br>( $1 - \alpha$ ) | II.Type Error<br>( $\beta$ )        |
| Reject $H_0$ | I.Type Error<br>( $\alpha$ )         | Correct Decision<br>( $1 - \beta$ ) |

$\alpha$  : Significance level. It is the probability of rejecting  $H_0$  when  $H_0$  is true.

$1 - \alpha$  : It states the confidence level of a test. It is the probability of accepting  $H_0$  when  $H_0$  is true.

$1 - \beta$  : The power of the test. It is the probability of rejecting  $H_0$  when  $H_0$  is false.

# Hypothesis Testing

## Statistics

### The Determination of $\beta$ Error for Population Mean

Consider the two-tailed hypothesis testing:

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

Assume that the population has normal distribution with known standard deviation  $\sigma$ . If  $H_0$  is true, then

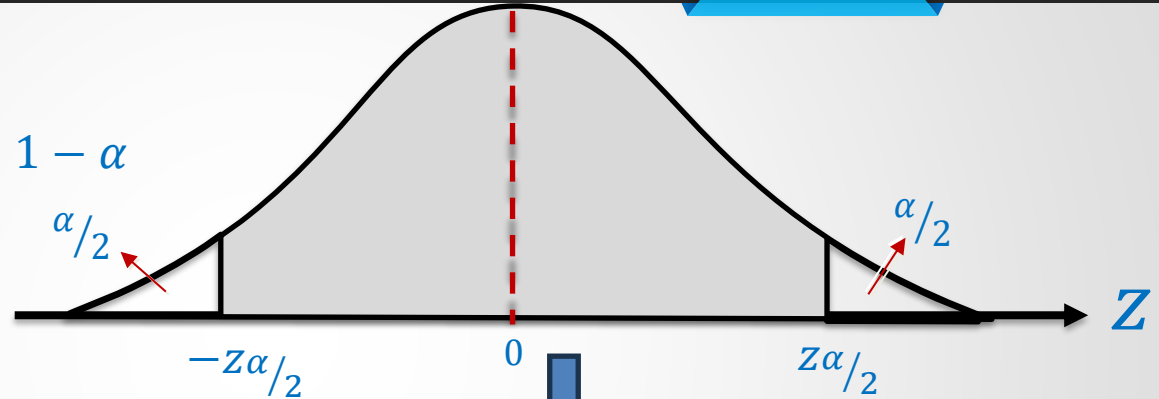
$$P\left(-Z_{\alpha/2} < \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < Z_{\alpha/2}\right) = 1 - \alpha$$

# Hypothesis Testing

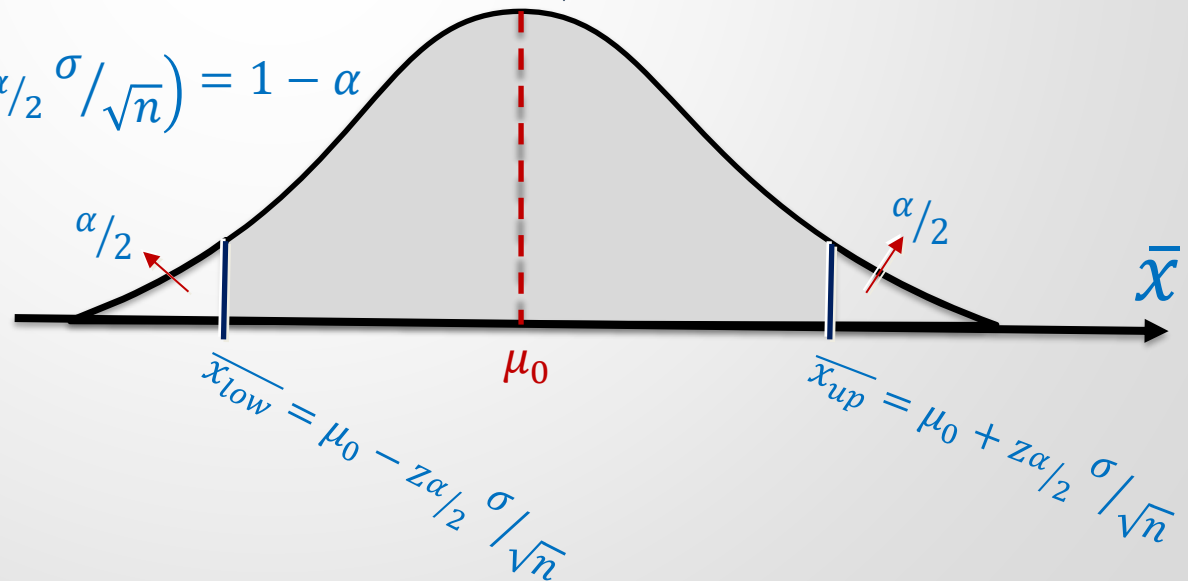
## Statistics

If  $H_0$  is true, then

$$P\left(-z_{\alpha/2} < \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$



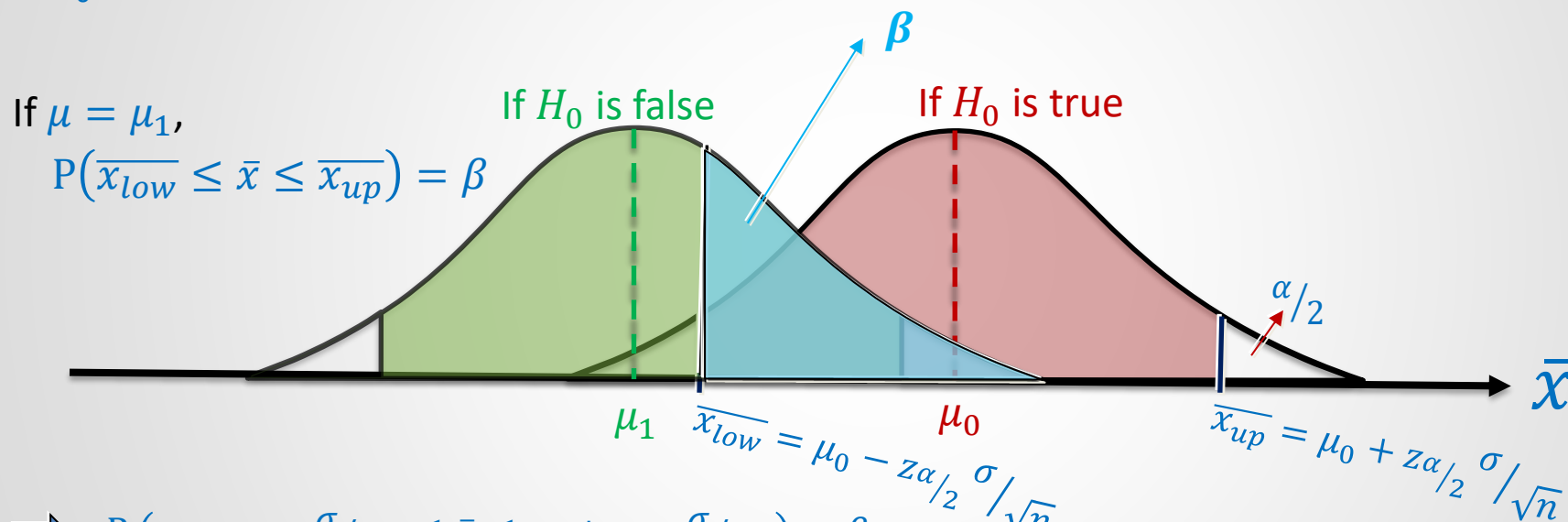
$$P\left(\mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{x} < \mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$



# Hypothesis Testing

## Statistics

If  $H_0$  is false and the exact value of the population mean is  $\mu_1$ , then ...



$$\Rightarrow P\left(\mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{x} \leq \mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = \beta$$

$$\Rightarrow P\left(\frac{\mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \mu_1}{\sigma/\sqrt{n}} \leq \frac{\bar{x} - \mu_1}{\sigma/\sqrt{n}} \leq \frac{\mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \mu_1}{\sigma/\sqrt{n}}\right) = \beta$$

$$\Rightarrow P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) = \beta$$

# Hypothesis Testing

## Statistics

If  $\beta$  is obtained similarly for one-tailed hypothesis, one can write the following probabilities for  $\beta$ :

➤ If  $H_0: \mu = \mu_0$   
 $H_a: \mu \neq \mu_0$   
$$P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) = \beta$$

➤ If  $H_0: \mu \geq \mu_0$   
 $H_a: \mu < \mu_0$   
$$P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha} \leq z\right) = \beta$$

➤ If  $H_0: \mu \leq \mu_0$   
 $H_a: \mu > \mu_0$   
$$P\left(z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha}\right) = \beta$$

# Hypothesis Testing

## Statistics



**Example 1:** A company produces a specific kind of soaps, and it claims that the soap's PH value is less than 5.5. 36 soaps are selected, and it is examined that the mean and standard deviation of their PH value is 5 and 1.5, respectively. Test the hypothesis for  $\alpha = 0.05$  and then find the power of test for  $\mu_1 = 5$ .

**The hypothesis for  $\alpha = 0.05$ :**

$$n = 36 \quad \bar{x} = 5 \quad s = 1.5$$

$$H_0: \mu = 5.5$$

$$H_a: \mu < 5.5$$

$$z_\alpha = z_{0.05} = 1.645$$

$$z_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{5 - 5.5}{1.5/\sqrt{36}} = -2$$

Since  $z_0 < -1.645$ ,  $H_0$  is rejected.

**The power of test for  $\mu_1 = 5$ :**  $1 - \beta = ?$

If we accept  $H_0$  when it is false, then

$$\Rightarrow P\left(\frac{\mu_0 - \mu_1}{s/\sqrt{n}} - z_\alpha \leq z\right) = \beta$$

$$\Rightarrow P\left(\frac{5.5 - 5}{1.5/\sqrt{36}} - 1.645 \leq z\right) = \beta$$

$$\Rightarrow P(0.36 \leq z) = \beta$$

$$\Rightarrow \beta = 0.5 - 0.1406 = 0.3594$$

$$\Rightarrow 1 - \beta = 0.6406$$

# Hypothesis Testing

## Statistics



**Example 2:** Solve the previous example by assuming that the population has a normal distribution with a known standard deviation of 1.5 for  $n=9$  and  $\mu_1 = 6$ .

**The hypothesis for  $\alpha = 0.05$ :**

$$n = 9 \quad \bar{x} = 5 \quad \sigma = 1.5$$

$$H_0: \mu = 5.5$$

$$H_a: \mu < 5.5$$

$$z_\alpha = z_{0.05} = 1.645$$

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{5 - 5.5}{1.5 / \sqrt{9}} = -1$$

Since  $z_0 > -1.645$ , we failed to reject  $H_0$ .

**The power of test for  $\mu_1 = 6$  :**

$$1 - \beta = ?$$

If we accept  $H_0$  when it is false, then

$$\Rightarrow P\left(\frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} - z_\alpha \leq z\right) = \beta$$

$$\Rightarrow P\left(\frac{5.5 - 6}{1.5 / \sqrt{9}} - 1.645 \leq z\right) = \beta$$

$$\Rightarrow P(-2.64 \leq z) = \beta$$

$$\Rightarrow \beta = 0.5 + 0.4959 = 0.9959$$

$$\Rightarrow 1 - \beta = 0.0041$$



# Hypothesis Testing

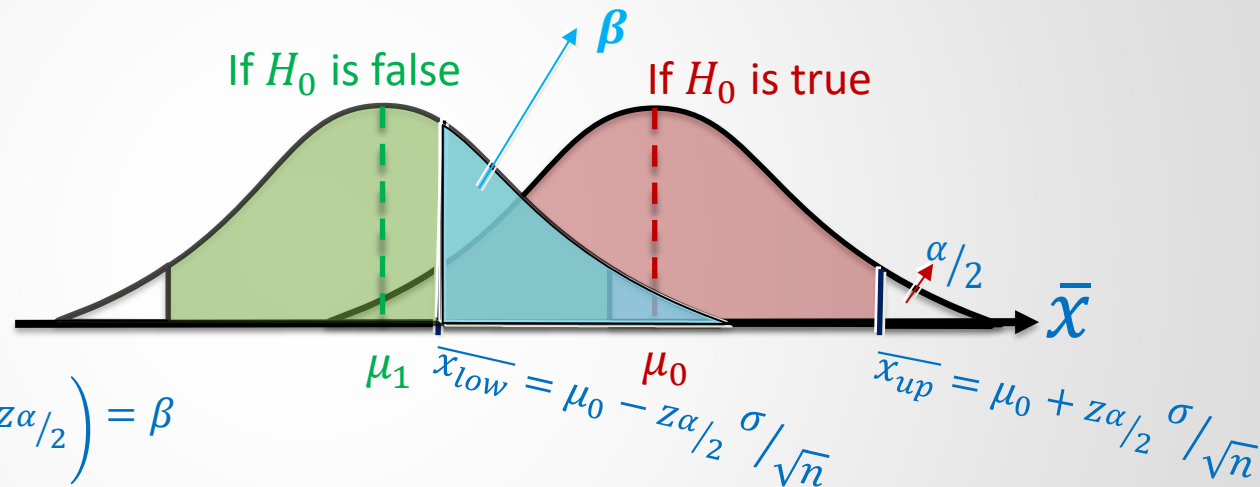
## Statistics

### DETERMINATION OF SAMPLE SIZE

For a two-tailed test,

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$



$$P\left(\frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} - z_{\alpha/2} \leq z \leq \frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} + z_{\alpha/2}\right) = \beta$$

If  $\mu_1 < \mu_0$

$$P(z \geq z_\beta) = \beta$$

If  $\mu_1 > \mu_0$

$$P(z \leq -z_\beta) = \beta$$

$$\left. \begin{aligned} \frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} - z_{\alpha/2} &= z_\beta \quad \text{and} \quad \frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} + z_{\alpha/2} = -z_\beta \end{aligned} \right\} n = \frac{(z_\beta + z_{\alpha/2})^2 \sigma^2}{(\mu_0 - \mu_1)^2}$$

# Hypothesis Testing

## Statistics

### DETERMINATION OF SAMPLE SIZE

For a one-tailed tests,

$$H_0: \mu \geq \mu_0$$

$$H_a: \mu < \mu_0$$

$$\downarrow$$
$$P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_\alpha \leq z\right) = \beta$$

$$\downarrow$$
$$P(z_\beta \geq z) = \beta$$

$$\downarrow$$
$$\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_\alpha = z_\beta$$

$$H_0: \mu \leq \mu_0$$

$$H_a: \mu > \mu_0$$

$$\downarrow$$
$$P\left(z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_\alpha\right) = \beta$$

$$\downarrow$$
$$P(z \leq -z_\beta) = \beta$$

$$\downarrow$$
$$\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_\alpha = -z_\beta$$

$$n = \frac{(z_\beta + z_\alpha)^2 \sigma^2}{(\mu_0 - \mu_1)^2}$$

# Hypothesis Testing

## Statistics



**Example 3:** Let  $\sigma=315$ ,  $\mu_0 = 2100$ ,  $\mu_1 = 2550$ ,  $\alpha=0.05$  and  $1-\beta=0.85$ . Find the sample size to satisfy the given conditions when applying a two-tailed test.

$$z_{\alpha/2} = z_{0.025} = 1.96$$

$$z_{\beta} = z_{0.15} = 1.04$$

$$n = \frac{(z_{\beta} + z_{\alpha/2})^2 \sigma^2}{(\mu_0 - \mu_1)^2} = \frac{(1.04 + 1.96)^2 315^2}{(2100 - 2550)^2} = 4.410$$

$$\Rightarrow n = [4.410] = 5$$